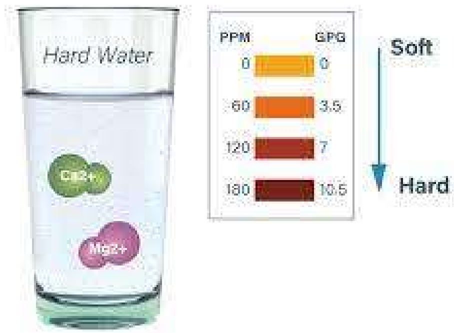
Unit 1: Water Chemistry

HARDNESS OF WATER

Hardness of water is defined as which prevents the lathering of soap. This is due to presence of certain salts like Ca^{+2}, Mg^{+2} and other heavy metals dissolved in water. Soaps (Sodium or Potassium salts of higher fatty acids) like Stearic acids ($C_{17}H_{35}COONa$).



Soft Water: The water which gives more lather with soap is called soft water.

$$\underset{\text{soap}}{C_{17}H_{35}COONa} + H_2O \rightarrow \underset{\text{Stearic acid}}{C_{17}H_{35}COOH} + NaOH$$

Hard Water: The water which does not give lather with soap is called hard water. This is due to presence of certain salts like Ca^{+2}, Mg^{+2} and other heavy metals dissolved in water

$$\underset{\text{soap (soluble)}}{2C_{17}H_{35}COONa} + \underset{\text{salts (soluble)}}{CaCl_2/MgCl_2} \rightarrow \underset{\text{insoluble scum}}{(C_{17}H_{35}COO)_2Ca/Mg} + 2NaCl$$

CAUSES OF HARDNESS

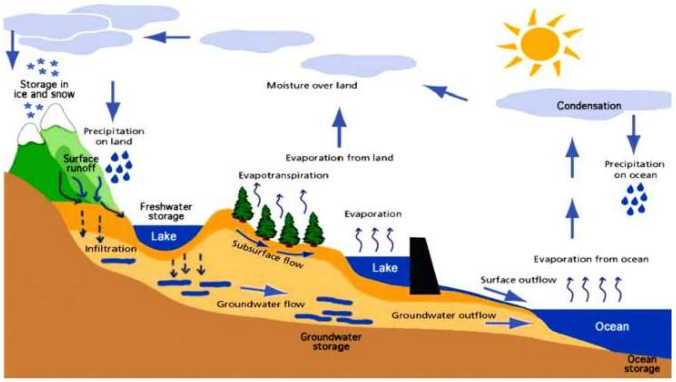
Hardness of water is due to the presence of Bicarbonates, Chlorides, Sulphates and Nitrates of Calcium and Magnesium. These soluble salts get mixed with natural water due to the following reasons:

1. When natural water containing CO_2 flows over the rocks of the limestone ($CaCO_3$) and Dolamite

   ($CaCO_3$ & $MgCO_3$), they get converted into soluble bicarbonates. Thus, water gets hardness.

$$\underset{\text{Insoluble}}{CaCO_3} + H_2O + CO_2 \rightarrow \underset{\text{Soluble}}{Ca(HCO_3)_2}$$

2. When natural water flows over the rocks containing chlorides and sulphates and Nitrates of Calcium and magnesium, these salts dissolve in water. Thus water gets hardness.



DISADVANTAGES OF HARDNESS

1. In Domestic use:

- **Washing:** Hard water, when used for washing purposes, does not producing lather freely with soap. As a result cleaning quality of soap is decreased and a lot of it is wasted.

- **Bathing:** Hard water does not lather freely with soap solution, but produces sticky scum on the bath-tub and body. Thus, the cleaning quality of soap is depressed and a lot of it is wasted.

- **Cooking:** The boiling point of water is increased because of presence of salts. Hence more fuel and time are required for cooking.

- **Drinking:** Hard water causes bad effects on our digestive system. Moreover, the possibility of forming calcium oxalate crystals in urinary tracks is increased.

2. Industrial Use:

- **Textile Industry:** Hard water causes wastage of soap. Precipitates of calcium and magnesium soaps adhere to the fabrics and cause problem.

- **Sugar Industry:** The water which containing sulphates, nitrates, alkali carbonates are used in sugar refining, cause difficulties in the crystallization of sugar.
- **Dyeing Industry:** The dissolved salts in hard water may reacts with costly dyes forming precipitates.
- **Paper Industry:** Calcium, magnesium, Iron salts in water may affect the quality of paper.
- **Pharmaceutical Industry:** Hard water may cause some undesirable products while preparation of pharmaceutical products.

3. Steam generation in Boilers:

- For steam generation, boilers are almost invariably employed. If the hard water is fed directly to the boilers, there arise many troubles such as: Scales & sludges formation, Corrosion, Priming & Foaming and Caustic embrittlement.

TYPES OF HARDNESS

Hardness of water is mainly two types:

1. Temporary Hardness 2. Permanent Hardness

1. Temporary or Carbonate Hardness: Temporary Hardness mainly caused by the presence of dissolved bicarbonates of Calcium, Magnesium (Ca $(HCO_3)_2$, Mg $(HCO_3)_2$). Temporary Hardness can be largely removed by boiling of water.

$$Ca(HCO_3)_2 \xrightarrow{\text{by heating}} CaCO_3\downarrow + H_2O + CO_2$$

CO_2 Calcium bicarbonate

$$Mg(HCO_3)_2 \xrightarrow{\text{by heating}} Mg(OH)_2\downarrow + CO_2$$

$2CO_2$ Magnesium bicarbonate

hard water soft water

2. Permanent or Non-Carbonate Hardness: It is due to the presence of dissolved Chlorides, Nitrates and Sulphates of Calcium, Magnesium, Iron and other metals. Permanent hardness responsible salts are $CaCl_2$, $MgCl_2$, $CaSO_4$, $MgSO_4$, $FeSO_4$, $Al_2(SO_4)_3$. Permanent Hardness cannot be removed by boiling but it can be removed by the use of ***chemical agents***.

EQUIVALENTS OF CALCIUM CARBONATE

The expression of hardness producing salts usually expressed in terms of an equivalent amount of $CaCO_3$. Calcium Carbonate is chosen as a standard because:

i. Its molecular weight (100) and equivalent weight (50) is a whole number, so the calculations in water analysis can be simplified.
ii. It is the most insoluble salt that can be precipitated in water treatment.

The conversion of the hardness causing salts into $CaCO_3$ equivalents can be achieved by using the following formula:

$$\text{The equivalent of CaCO3} = \frac{\text{The weight of hardness causing salts}}{\text{Molecular weight of hardness causing salts}} \times 100 \text{ (Molecular weight of } CaCO_3\text{)}$$

Table: Calculations of equivalents of Calcium Carbonate ($CaCO_3$)

Dissolved salt/ion	*Molar mass*	*Chemical equivalent*	*Multiplication factor for conveting into equivalents of $CaCO_3$*
$Ca(HCO_3)_2$	162	81	100/162
$Mg(HCO_3)_2$	146	73	100/146
$CaSO_4$	136	68	100/136
$CaCl_2$	111	55.5	100/111
$MgSO_4$	120	60	100/120
$MgCl_2$	95	47.5	100/95
$CaCO_3$	100	50	100/100
$MgCO_3$	84	42	100/84
CO_2	44	22	100/44
$Ca(NO_3)_2$	164	82	100/164
$Mg(NO_3)_2$	148	74	100/148

UNITS OF HARDNESS

1. **Parts per Million (ppm):** The number of parts of calcium carbonate equivalent hardness presents in 10^6 parts of water.

$$1ppm = 1 \text{ part of } CaCO_3 \text{ eq hardness in } 10^6 \text{ parts of water.}$$

2. **Milligrams per litre (mg/l):** The number of milligrams of calcium carbonate equivalent hardness presents in litre of water.

$$1 \text{ mg/L} = 1 \text{ mg of } CaCO_3 \text{ eq hardness in 1 litre of water.}$$

But one litre of water weights =1 kg =1000g = 1000 x 1000 mg = 10^6 mg = 1 ppm.

3. **Clark's degree (°Cl):** The number of parts of calcium carbonate equivalent hardness presents in 70,000 or (7×10^4) parts of water.

$$1° \text{ Clarke} = 1 \text{ part of } CaCO_3 \text{ eq hardness per 70,000 parts of water.}$$

4. **Degree French (°Fr):** The number of parts of calcium carbonate equivalent hardness presents in 10^5 parts of water.

$$1° \text{ Fr} = 1 \text{ part of } CaCO_3 \text{ hardness eq per } 10^5 \text{ parts of water.}$$

Relationship between various units of hardness:

1 ppm	= 1 mg/L	= 0.1° Fr	= 0.07° Cl
1 mg/L	= 1 ppm	= 0.1° Fr	= 0.07° Cl
1 ° Cl	= 1.433° Fr	= 14.3 ppm	= 14.3 mg/L
1 ° Fr	= 10 ppm	= 10 mg/L	= 0.7° Cl

Numerical Problems based on the hardness of water:

Problem-1: A sample of water is found to contains following dissolving salts in milligrams per litre $Mg(HCO_3)_2$ = 73, $CaCl_2$ = 111, $Ca(HCO_3)_2$ = 81, $MgSO_4$ = 40 and $MgCl_2$ = 95. Calculate temporary and permanent hardness and total hardness.

Solution:

Name of the hardness causing salts	Amount of the hardness causing salts(mg/Lit)	Molecular weight of hardness causing salts	Amounts equivalent to $CaCO_3$ (mg/Lit)
$Mg(HCO_3)_2$	73	146	$73 \times 100/146 = 50$
$CaCl_2$	111	111	$111 \times 100/111 = 100$
$Ca(HCO_3)_2$	81	162	$81 \times 100/162 = 50$
$MgSO_4$	40	120	$40 \times 100/120 = 33.3$
$MgCl_2$	95	95	$95 \times 100/95 = 100$

Temporary hardness = $Mg(HCO_3)_2 + Ca(HCO_3)_2$
= 50 + 50 = 100mgs/Lit.

Permanent hardness = $CaCl_2 + MgSO_4 + MgCl_2$
= 100 + 33.3 + 100 = 233.3mgs/Lit.

Total hardness = Temporary hardness + Permanent hardness
= 100 + 233.3 = 333.3mgs/Lit.

Problem-2: A sample of water is found to contains following dissolving salts in milligrams per litre $Mg(HCO_3)_2$ = 16.8, $MgCl_2$ = 12.0, $MgSO_4$ = 29.6 and NaCl = 5.0. Calculate temporary and permanent hardness of water.

Solution:

Name of the hardness causing salts	Amount of the hardness causing salts(mg/Lit)	Molecular weight of hardness causing salts	Amounts equivalent to $CaCO_3$ (mg/Lit)
$Mg(HCO_3)_2$	16.8	146	$16.8 \times 100/146 = 11.50$
$MgCl_2$	12.0	95	$12.0 \times 100/95 = 12.63$
$MgSO_4$	29.6	120	$29.6 \times 100/120 = 24.66$
NaCl	5.0	NaCl does not contribute any hardness to water hence it is ignored.	

Temporary hardness = $Mg(HCO_3)_2$ = 11.50mgs/Lit.

Permanent hardness = $MgCl_2 + MgSO_4$ = 12.63 + 24.66 = 37.29mgs/Lit.

Problem-3: A sample of water is found to contains following analytical data in milligrams per litre $Mg(HCO_3)_2$ = 14.6, $MgCl_2$ = 9.5, $MgSO_4$ = 6.0 and $Ca(HCO_3)_2$ = 16.2. Calculate temporary and permanent hardness of water in parts per million, Degree Clarke's and Degree French.

Solution:

Name of the hardness causing salts	Amount of the hardness causing salts(mg/Lit)	Molecular weight of hardness causing salts	Amounts equivalent to $CaCO_3$ (mg/Lit)
$Mg(HCO_3)_2$	14.6	146	$14.6\times100/146 = 10$
$MgCl_2$	9.5	95	$9.5\times100/95 = 10$
$MgSO_4$	6.0	120	$6.0\times100/120 = 5$
$Ca(HCO_3)_2$	16.2	162	$16.2\times100/162 = 10$

Temporary hardness $[Mg(HCO_3)_2 + Ca(HCO_3)_2] = 10 + 10 = 20$mg/Lit

$= 20$ppm

$= 20\times0.07°Cl = 1.4°Cl$

$= 20\times0.1°Fr = 2°Fr$

Permanent hardness $[MgCl_2 + MgSO_4] = 10 + 5 = 15$mg/Lit

$= 15$ppm

$= 15\times0.07°Cl = 1.05°Cl$

$= 15\times0.1°Fr = 1.5°Fr$

Types of Impurities present in the water or characteristics imparted by impurities in water

Three types:

1. **Physical impurities**
2. **Chemical impurities**
3. **Biological impurities**

1. Physical impurities:

a) **Colour:** The colour in water is caused by metallic substances like salts of **iron, manganese, industrial effluents**. This is due to dissolved substances and substances present as fine colloids.
For example, the **yellowish-red colour** indicates the presence of **iron**.

b) **Turbidity:** is due to the colloidal, extremely fine suspensions such as insoluble substances like **clay, slit,** and **micro-organisms**.
Turbidity in water can be eliminated by **sedimentation, coagulation, filtration**, etc.

c) **Taste:** The presence of dissolved minerals in water produces taste.
The **bitter taste** can be due to the presence of **Fe, Al, Mn, Sulphates, and lime**.
Soap taste can be due to the presence of large amount of **sodium bicarbonate**.

d) **Odour:** Disagreeable odor in water may be caused by the presence of living organisms, and decaying vegetation, including algae, bacteria, and fungi.

Types of Impurities present in the water or characteristics imparted by impurities in water

2. **Chemical impurities:** **It includes:**

- **Inorganic** and **organic chemicals** released from dyes, paints, vanishes, drugs, insecticides, pesticides, etc. All these pollute water bodies.
- **Acids discharged** in water by **DDT, high explosives, battery, industries**, etc.

The use of this type of contaminated water cause harmful effects on health of human-beings.

3. **Biological Impurities:**

- Biological impurities are **algae, pathogenic bacteria, fungi, viruses, pathogens**, **parasite-worms**, etc.
- The sources of these contamination is **discharge of domestic and sewage wastes**, **excreta** (from man, animals and birds), etc.

BOILER TROUBLES/PROBLEAMS

A boiler is a closed vessel in which water under pressure is transformed into steam by the application of heat. The steam so generated is used in industries and generation of power. In modern pressure boilers and laboratories, the water required is used pure than the distilled water.

A boiler feed water should correspond with the following composition:

- Its hardness should be below 0.2ppm.
- Its caustic alkalinity (due to OH^-) should lie between 0.15ppm to 0.45ppm.
- It's should be free from dissolved gases like O_2, CO_2, in order to prevent boiler corrosion.

Excess of impurities in the boiler feed water generally cause the following problems:

1. **Sludge's and Scale formation**
2. **Priming and Foaming**
3. **Caustic embrittlement**
4. **Boiler Corrosion**

1. SLUDGES AND SCALES FORMATION

SLUDGES:

- Sludge is a soft, loose, and slimy precipitate formed within the boiler.
- It is formed at comparatively **colder portions** of the boiler and collects in the area where **flow rate is slow**.
- Ex: $MgCO_3$, $MgCl_2$, $CaCl_2$, $MgSO_4$.

Reasons for formation of sludges:

- The dissolved salts whose solubility is more in hot water and less in cold water produce sludges.

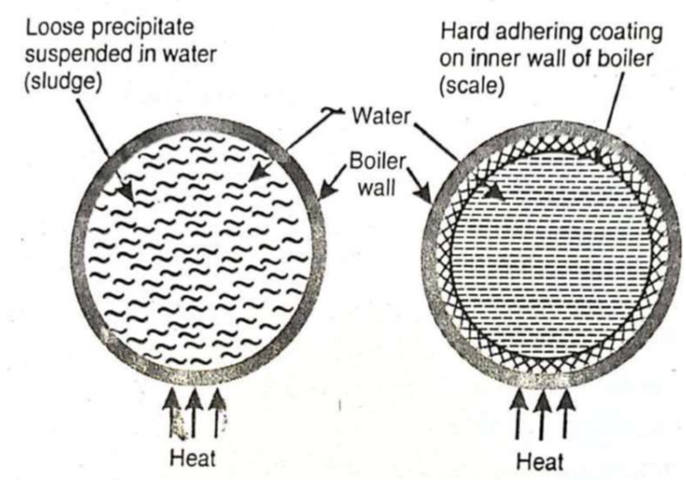


Fig. Scale and Sludge in boilers

Disadvantages of sludges:

- Sludges are bad conductors of heat and results in the wastage of heat and fuel.
- Excessive sludge formation leads to the **settling of sludge in slow circulation areas** such as pipe connections, plug openings, gauge–glass connections, leading to the choking of the pipes.

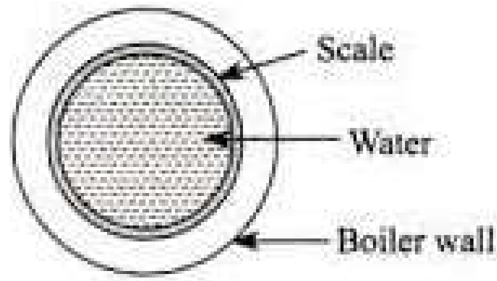
Prevention of sludge formation:

- Using soft water, which is free from dissolved salts like MgCO3, MgCl2, CaCl2, and MgSO4, can prevent sludge formation.
- By blow down operation carried out frequently can prevent sludge formation.

SCALES:

- Scales are hard, adhering precipitates formed on the inner walls of the boilers.
- Scales are stick very firmly on to the inner walls of the boiler.

  - **formed by the substances like $Ca(HCO_3)_2$, $CaSO_4$ and $Mg(OH)_2$.**



(b) Scale in boiler

Reasons for formation of scales:

a. **Decomposition of calcium bicarbonate:** The calcium bicarbonate at high temperature decomposes to calcium carbonate which is insoluble salt, forms scale in low pressure boilers.

$$Ca(HCO_3)_2 \longrightarrow \underset{(Scale)}{CaCO_3\downarrow} + H_2O + CO_2\uparrow$$

$$CaCO_3 + H_2O \longrightarrow \underset{(Sludge)}{Ca(OH)_2} + CO_2\uparrow$$

b. **Hydrolysis of Magnesium salts:** Magnesium salts gets hydrolyzed at high temperature forming $Mg(OH)_2$ precipitation which forms salt type scale.

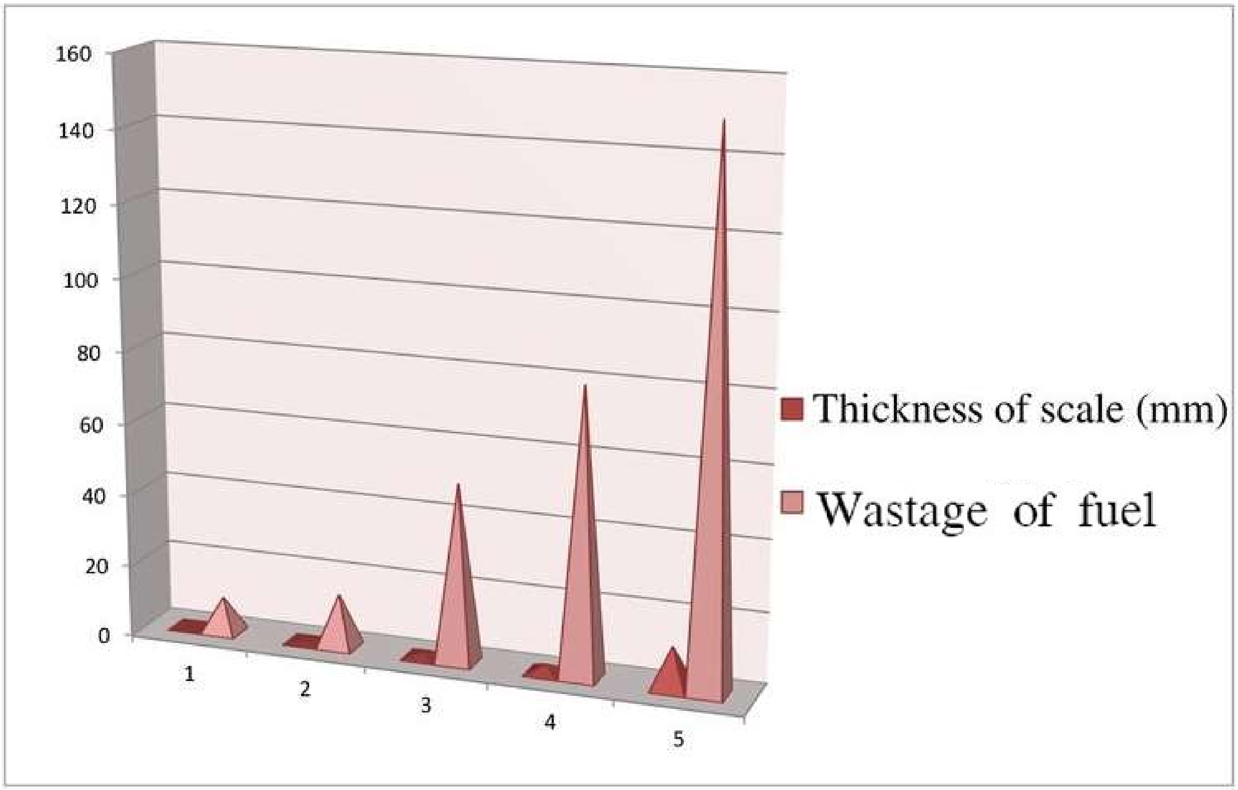
$$MgCl_2 + H_2O \rightarrow Mg(OH)_2\downarrow + 2HCl$$

Disadvantages of Scales:

1. **Wastage of heat and fuels:** Scales have poor thermal conductivity, so the rate of heat transformation is reduced.
2. **Lowering of boiler safety** is due to overheating of the boiler material becoming softer and weaker, which causes distortion of the boiler.
3. **Decrease in efficiency** of the boiler due to scales deposited in the values and condensers of the boiler cause choking.
4. **Danger of explosion,** which happens with the formation of the scales, the boiler plate faces higher temperature outside and lesser temperature inside due to uneven expansion.

Prevention of scales:

1. By giving thermal shocks, by sudden heating and sudden cooling which makes scale brittle and removed by scrubbing with wire brush.
2. If the scale is very hard, that is formed by $CaCO_3$ can be removed by washing with 5-10% HCl, and $CaSO_4$ can be removed with EDTA solution.

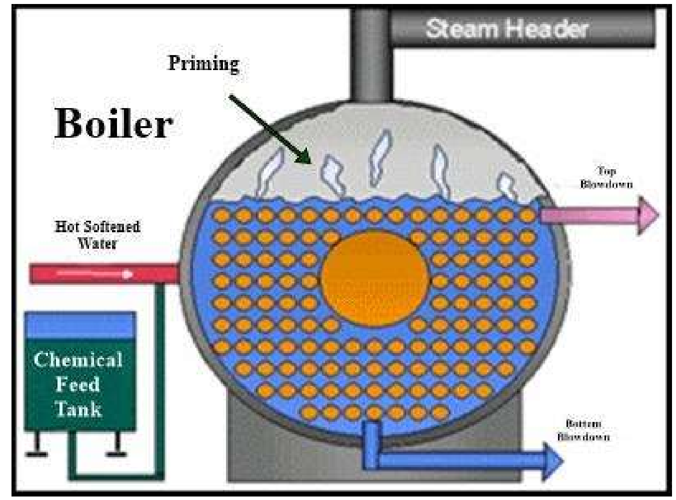


Thickness of scale (mm)	0.325	0.625	1.25	2.5	12
Wastage of fuel	10%	15%	50%	80%	150%

2. PRIMING AND FOAMING

PRIMING:

When a boiler is steaming (i.e., producing steam) rapidly, some particles of the liquid water are carried along with the steam. This process of ***'wet steam'*** formation is known as **'Priming'**.



Causes:

Priming is caused by:

1. The presence of large amount of dissolved salts
2. High steam velocities
3. Sudden boiling
4. Improper boiler design
5. Sudden increase in steam production rate.

Prevention:

1. By controlling the velocity of steam
2. Maintaining low water levels in boilers
3. Effective softening and filtration of boiler feed water
3. Using good boiler design

FOAMING:

The phenomenon of formation of persistent foam or bubbles on the surface of water inside the boiler, which do not break easily.

Causes:

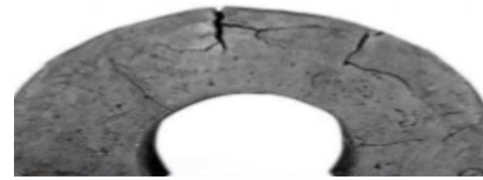
1. Water containing dissolved impurities and suspended matter has a greater tendency to produce foam.
2. The presence of a large quantity of suspended impurities and oils lowers the surface tension, producing foam.

Prevention:

1. Adding antifoaming chemicals like castor oil or polyamides.
2. Besides castor oil, Gallic acid and tannic acids, corn oil, cotton seed oil, sperm oil, bees wax, etc., are also used as antifoaming agents.
3. Blow down operation of the boiler can prevent foaming.
4. Removing oil form boiler water by adding compounds like sodium aluminate.

3. CAUSTIC EMBRITTLEMENT

The formation of brittle and in crystalline cracks in the boiler shell is called caustic embrittlement.

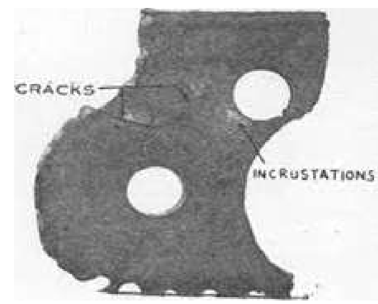


- The main reason for this is the presence of **alkali-metal carbonates** and **bicarbonates** in feed water.
- In lime-soda process, some residual **Na_2CO_3** is still present in the softened water.
- This Na_2CO_3 decomposes to give **NaOH and CO_2**, due to which the boiler water becomes "Caustic Soda".

$$Na_2CO_3 + H_2O \rightarrow 2NaOH + CO_2$$

- The H_2O evaporates, the concentration of NaOH increase progressively creating a **concentration cell** as given below thus **dissolving the iron of the boiler** as sodium ferrate (Na_2FeO_2).
- Caustic attack on boiler parts can be represented as:

+			−
Iron at rivettes, bends, Joints etc.	Conc. NaOH soln.	Dil. NaOH soln.	Iron at Plane surfaces



- The iron at plane surfaces surrounded by dilute NaOH becomes **cathodic** while the iron at bends and joints surrounded by highly concentrated NaOH becomes **anodic** which consequently decayed or corroded.
- This causes embrittlement of boiler parts such as bends, joints, reverts etc, due to which the **boiler gets fail**.

Caustic embrittlement can be prevented:

- By maintaining the **pH** value of water and **neutralization of alkali**.
- By using **Sodium Phosphate** as softening reagents, in the external treatment of boilers.
- It can also be prevented by adding **Tannin** or **Lignin** or **Sodium sulphate** which prevents the infiltration of caustic-soda solution blocking the hair-cracks.

4. BOILER CORROSION

Boiler corrosion:
The decay of boiler material by chemical or electrochemical attack by its environment is known as "Boiler corrosion".

The main reasons for boiler corrosion are:

1. Dissolved Oxygen: water contains usually about 8ml of dissolved oxygen per liter at room temperature.

Disadvantages of dissolved oxygen:

In presence of prevailing high temperature, the dissolved Oxygen attacks boiler material as follows:

$$2Fe + 2H_2O + O_2 \longrightarrow 2Fe(OH)_2\downarrow$$

$$4Fe(OH)_2\downarrow + O_2 \longrightarrow 2[Fe_2O_3.2H_2O]\downarrow \text{ (rust)}$$

Removal of dissolved Oxygen:

1)By adding calculated quantity of Sodium sulphite (Na_2SO_3) or hydrazine(N_2H_4) or sodiumsulphide (Na_2S).

$$2Na_2SO_3 + O_2 \longrightarrow 2\,Na_2SO_4$$

$$N_2H_4 + O_2 \longrightarrow N_2 + 2H_2O$$

$$Na_2S + 2O_2 \longrightarrow Na_2SO_4$$

Hydrazine is an ideal internal treatment chemical for the removal of dissolved oxygen.
Azamina 8001-RD (a polyvalent organic compound) has been employed for degassing water in minimum time.

2. Dissolved Carbon dioxide: has a slow corrosive effect on the materials of boiler plate. Under the high temperature and pressure the bicarbonates decompose to produce CO2.

$$Mg(HCO_3)_2 \xrightarrow{\Delta} MgCO_3 + H_2O + CO_2\uparrow$$

Disadvantages of CO_2:

It has slow corrosive effect on boiler plates by producing the carbonic acid.

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

Removal of CO_2:

By adding calculated amount of ammonia.

$2NH_4OH + CO_2 \rightarrow (NH_4)_2CO_3 + H_2O$

By mechanical deaeration process along with oxygen.

3. *Acids from dissolved salts:* *water* containing dissolved Mg salts liberate acids on hydrolysis.

Ex: $MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 \downarrow + 2HCl$

Disadvantages of acids:

The acids react with the iron of boiler plate in a chain reaction producing HCl again and again.

$Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$

$FeCl_2 + 2H_2O \rightarrow Fe(OH)_2\downarrow + 2HCl$

Presence of even a small amount of $MgCl_2$ will have corrosion of iron to a large extent.

Prevention of acid corrosion:

1) By softening of boiler water to remove $MgCl_2$ from the water.
2) By frequent blow down operation.
3) Addition of inhibitors to form a thin film on the surface of the boiler protecting the metal from attack.

Sodium silicates, sodium phosphates and sodium chromate acts as good corrosion inhibitors.

Softening of Hard Water

External conditioning / treatment

- Lime-soda process
- Zeolite process
- Ion exchange process

Internal conditioning / treatment / Sequestration

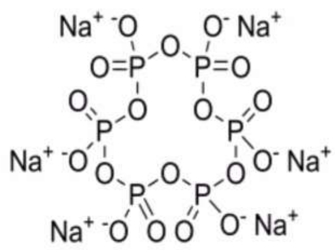
- Carbonate conditioning
- Phosphate conditing
- Calgon conditioning
- Collidal conditioning

INTERNAL TREATMENT OF WATER

Suitable **chemicals** are added to the boiler water either to precipitate or to convert the scale into compounds is called internal treatment of the boiler feed water.

1. Calgon conditioning
2. Phosphate conditioning
3. Colloidal conditioning
4. Carbonate conditioning

Sodium Hexametaphosphate



Calgon

1. Calgon conditioning:

- Involves in adding calgon to boiler water.
- **Calgon** = **Sodium hexa meta phosphate** = $Na_2[Na_4(PO_3)_6]$
- The addition of Calgon to hard water causes the calcium and magnesium ions of hard water to displace sodium ions from the anion of Calgon.
- This results in the removal of calcium and magnesium ions from hard water in the form of a complex with Calgon.

$$Na_2[Na_4(PO_3)_6] + 2Ca^{2+} \Rightarrow Na_2[Ca_2(PO_3)_6]$$

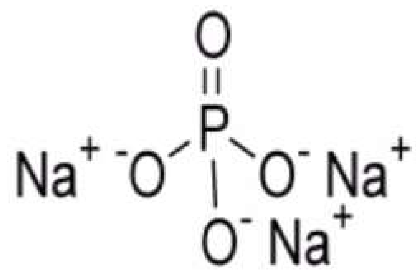
CALGON CALCIUM ION SOLUBLE COMPLEX

$$Na_2[Na_4(PO_3)_6] \rightarrow 2Na^+ + [Na_4P_6O_{18}]^{-2}$$

$$2CaSO_4 + [Na_4P_6O_{18}]^{-2} \rightarrow [Ca_2P_6O_{18}]^{-2} + 2Na_2SO_4$$

2. Phosphate conditioning:

Trisodium Phosphate



- The addition of **Trisodium phosphate (Na_3PO_4)** in hard water.
- It reacts with the hardness causing agents and gives **calcium and magnesium phosphates** which are soft and nonadhere and can be removed easily by **blow-down operation**.

$$3CaCl_2 + 2Na_3PO_4 \rightarrow Ca_3(PO_4)_2\downarrow + 6NaCl$$

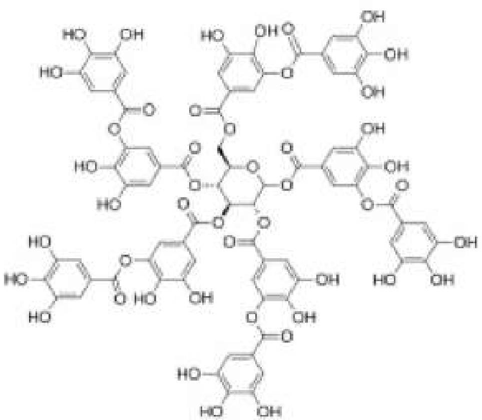
$$3MgCl2 + 2Na3PO4 \rightarrow Mg_3(PO\}_4\downarrow + 6NaSO_4$$

Three types of Phosphates are employed.

i. **Tri sodium Phosphate (Na_3PO_4):** is too alkaline used for treat to too acidic water.
ii. **Di sodium Phosphate (Na_2HPO_4):** is weakly alkaline used for treat to weakly acidic water.
iii. **Sodium dihydrogen Phosphate (NaH_2PO_4):** is too acidic used for treat to too alkaline water.

3. Colloidal conditioning:

- In low-pressure boilers, scale formation can be avoided by adding organic substances like kerosene, tannin, agar-agar (a gel), etc.,
- These substances get **coated** over the scale forming precipitates, thereby **yielding non-sticky and loose deposits**, which can easily be removed by using blow-down operation.



TANNIN CHEMICAL STRUCTURE

AGAR-AGAR IN ITS GEL LIKE STRUCTURE

4. Carbonate conditioning:

- Reagent added is **sodium carbonate (Na_2CO_3)**.

- In low pressure boilers scale formation can be avoided by adding sodium carbonate to boiler water.

$$CaSO4 + Na2CO3 \rightarrow \underset{\text{Sludge}}{CaCO3\downarrow} + Na2SO4$$

  - Deposition of $CaSO_4$ as scale doesn't takes place and **calcium is precipitated as loose sludge of $CaCO_3$** which can be removed by blow down operation.

EXTERNAL TREATMENT:

In industry, four external methods are mainly employed for softening of water.

1. ZEOLITE OR PERMUTIT PROCESS

2. LIME-SODA PROCESS

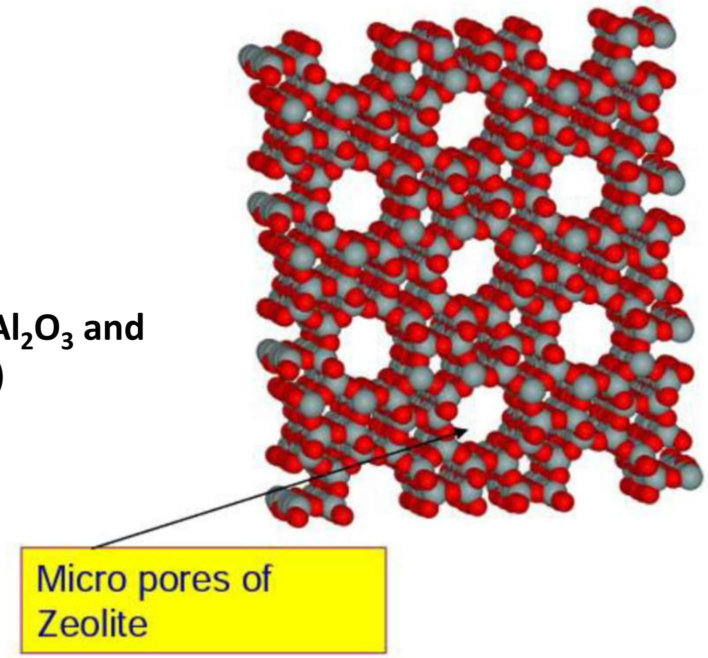
3. ION EXCHANGE/DEMINERALIZATION PROCESS

4. DESALINATION OF BRACKISH WATER–REVERSE OSMOSIS (RO)

1. ZEOLITE OR PERMUTIT PROCESS

- Zeolites are **hydrated sodium aluminum silicates $Na_2O.Al_2O_3$ $xSiO_2.yH_2O$**. (x=2-10 and y=2-6) (inorganic salts).

- They work as **water softeners** by replacing the **calcium and magnesium ions** in water with the **Sodium ions** in the zeolite.



Porous Structure of zeolite

1. **Natural zeolites:** Non-porous, amorphous and durable
   e.g. Na2O. Al2O3.3SiO2.2H2O

2. **Synthetic zeolites:** Gel like structures, generally porous **Na_2CO3, Al_2O_3 and SiO_2** heated together **$Na_2O.Al_2O3$ $xSiO2.yH_2O$**. (x=2-10 and y=2-6)
   e.g. Silicalite-1 and ZSM-5 (MFI)

- In this process hard water is allowed to pass through a **bed of zeolite** at a specified rate.
- Then the **sodium ions** present in the zeolite bed continuously replace the **calcium and magnesium ions** present in water and hence the water becomes soft.

$$Na_2Ze + Ca(HCO_3)_2 \rightarrow CaZe + 2NaHCO_3$$

$$Na_2Ze + Mg(HCO_3)_2 \rightarrow MgZe + 2NaHCO_3$$

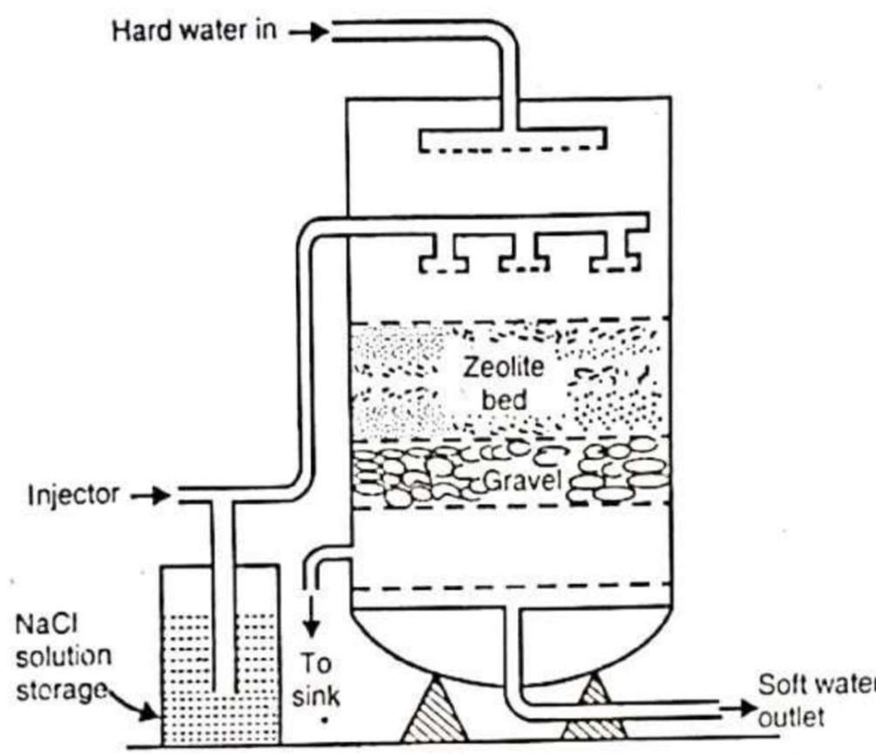


Fig.: Zeolite Softner

Regeneration:

- When the zeolite bed becomes exhausted it requires regeneration.
- This is achieved by passing 10% NaCl solution through it.

$$CaZe + 2NaCl \rightarrow Na_2Ze + CaCl_2$$

Advantages:

1. Almost complete removal of hardness **(10ppm)**

2. It is **compact**

3. Requires only **less time for softening**

4. **No sludge** formation since no precipitate is formed

5. Can **work under pressure**.

Disadvantages:

1. **More sodium salt** concentration in softened water.

2. **Turbidity** containing water cannot be used

3. Process exchange **only Ca^{2+} and Mg^{2+}** ions but cannot exchange **HCO_3^- and CO_3^{2-} ions**. So, it cannot be used in boilers.

4. If **Fe^{2+} and Mn^{2+}** are present in large quantities. They form respective zeolites, so **zeolites cannot be regenerated**

5. Water consisting of **high alkalinity or acidity** cannot be used because **zeolite is decomposed**

Numerical Problems based on Zeolite Process

Regeneration of NaCl Solution

- **Concentration of NaCl (Strength, Normality, W/V %)**
- **Volume of NaCl Solution**

Hard Water

- Hardness
- Volume of Water

Q1. Determination of Hardness
Q2. Determine the volume of water
Q3. Determine the volume of NaCl solution required for regeneration
Q4. Determine the concentration of NaCl solution.

Example 1 . *The hardness of 50,000 litres of a sample was removed by passing it through a zeolite softener. The softener then required 200 litres of NaCl solution, containing 125 g/litre of NaCl for regeneration. Calculate the hardness of the sample of water.*

Solution. 200 L of NaCl solution

$\equiv 200 \text{ L} \times 125 \text{ g/L} = 25{,}000 \text{ g NaCl} = (25{,}000 \times 50)/58{\cdot}5 \text{ g } CaCO_3 \text{ eq.}$

$= 21{,}368 \text{ g } CaCO_3 \text{ eq.}$

$\therefore$ Hardness of 50,000 L water $\equiv 21{,}368 \text{ g } CaCO_3$ eq.

or hardness of 1 L water $\equiv \frac{21{,}368}{50{,}000} = 0.4274 \text{ g } CaCO_3$ eq.

$= 427.4 \text{ mg } CaCO_3$ eq.

Hence, hardness of water $=$ **427.4 mg/L (or ppm).**

Example 2. *An exhausted Zeolite softener was regenerated by passing 150 litres of NaCl solution, having a strength of 150 g/l of NaCl. How many litres of hard water sample, having hardness of 600 ppm can be softened, using this softener?*

Solution.

150 litres of NaCl solution contains

$$150 \times 150 \text{ g} = 22{,}500 \text{ g Na Cl}$$

$$\equiv 22{,}500 \times \frac{50}{58.5} \text{ g of } CaCO_3 \text{ equivalent hardness.}$$

Given that 1 litre of hard water contains 600 ppm hardness $\equiv$ 600 mg of $CaCO_3$ = 0.6 g of $CaCO_3$

$\therefore$ The amount of hard water that can be softened by this softener

$$= \frac{22{,}500 \times 50}{0.6 \times 58.5} = \mathbf{32{,}051 \text{ litres.}}$$

Example 3. *The hardness of 1000 litres of a water sample was completely removed by a Zeolite softener. The Zeolite had required 30 litres of NaCl solution, containing 1,500 mg/l of NaCl for regeneration. Calculate the hardness of the water sample.*

Solution.

30 litres of NaCl solution contains

$$= 1.5 \times 30 \text{ g} = 45 \text{ g of NaCl.}$$

$$\equiv 45 \times \frac{50}{58.5} \text{ g of } CaCO_3 \text{ equivalent hardness.}$$

This much hardness may be deemed to be present in 1000 litres of water sample.

$\therefore$ Hardness of the water sample

$$= 45 \times \frac{50}{58.5} \times \frac{1,000}{1,000} \text{ mg/l}$$

$$= \mathbf{38.46 \text{ ppm.}}$$

Numerical Problems for Practice:

1. The hardness of 100,000 litres of a sample of water was completely removed by passing it through a zeolite softener. The softener then required 400 litres of sodium chloride solution containing 100 g/litre of NaCl for regeneration. Calculate the hardness of the water, sample.

2. A zeolite softener was 90% exhausted, when 10,000 L of hard water was passed through it. The softener required 200 L of NaCl solution of strength 50 g NaCl/L of solution. What is the hardness of water ?

3. An exhausted zeolite softener was regenerated by pasing 150 litres of NaCl solution, having a strength of 150 g/L of NaCl. If the hardness of water is 600 ppm, calculate the total volume of water that is softened by this softener.

2. LIME-SODA PROCESS

Principle: The lime soda process involves the **chemical conversion** of all the soluble hardness causing salts by the addition of **soda (Na_2CO_3)** and **lime [$Ca(OH)_2$]** into **insoluble precipitates** which could easily be removed by **settling and filtration**.

Functions of lime Removes:

1. Temporary hardness
2. Permanent Mg hardness
3. Dissolved Fe, Al salts
4. Dissolved CO_2 and H_2S gases
5. Free mineral acids Present in water

Lime:

The **temporary hardness** of water can be removed by hydrated lime. It react with $Ca(HCO_3)_2$ & $Mg(HCO_3)_2$ to form **insoluble precipitate of calcium carbonate and magnesium hydroxide** respectively.

$$Ca(HCO_3)_2 + Ca(OH)_2 \longrightarrow 2CaCO_3\downarrow + 2H_2O + CO2$$

$$Mg(HCO_3)_2 + 2Ca(OH)_2 \longrightarrow Mg(OH)_2\downarrow + 2CaCO_3\downarrow + 2H_2O$$

Hydrated lime is also used to remove **permanent hardness** (**magnesium salt impurities only**) from water. It reacts with $MgSO_4$ & $MgCl_2$ to form **insoluble precipitate of magnesium hydroxide.**

$$MgCl_2 + Ca(OH)_2 \longrightarrow Mg(OH)_2\downarrow + CaCl_2$$

$$MgSO_4 + Ca(OH)_2 \longrightarrow Mg(OH)_2\downarrow + CaSO_4$$

In above reaction calcium based impurities like $CaCl_2$ & $CaSO_4$ also form, which are soluble in water. Therefore **calcium based impurities** ($CaCl_2$ & $CaSO_4$) are not removed by lime treatment.

Soda:

Sodium carbonate (Na_2CO_3) is used to remove **permanent hardness** of water which caused
By $MgSO_4$ & $MgCl_2$ or $CaCl_2$ & $CaSO_4$.

$$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3\downarrow + 2NaCl$$
$$CaSO_4 + Na_2CO_3 \longrightarrow CaCO_3\downarrow + Na_2SO_4$$
$$MgCl_2 + Na_2CO_3 \longrightarrow MgCO_3\downarrow + 2NaCl$$
$$MgSO_4 + Na_2CO_3 \longrightarrow MgCO_3\downarrow + Na_2SO_4$$

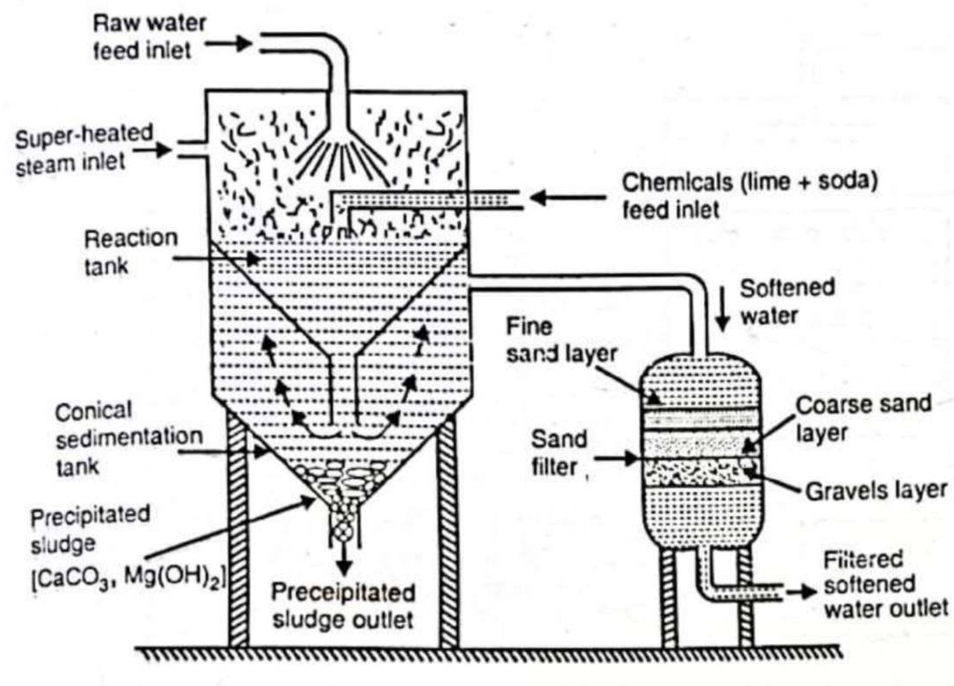


Fig. Continuous hot lime-soda softner

It consists of

- a reaction tank
- a conical sedimentation vessel and
- a sand filter.

Process

- The lime-soda process can be carried out both at room temperature as well as at higher temperatures.
- The process carried out at **room temperature** is called **cold lime–soda process** and that carried out at 80°-150°C is called **hot lime-soda process**.

Hot Lime-Soda Process

- Water is treated with the softening chemicals at a temperature of 80°-150°C.
- Since the process is carried out at a temperature close to the boiling point of the solution, the **reaction proceeds faster** and the **softening capacity of the process increases** several times.

Advantages of high temperature:

- The reaction proceeds faster.
- Softening capacity is increased.
- No coagulant needed as the precipitate and sludge formed settle down rapidly.
- Much of the dissolved gases are driven out of water.
- Viscosity of soften water is lower, so filtration of water becomes easier.
- This process produces water of comparatively low residual hardness 15 to 30 ppm.

Cold Lime Soda Process:

- Water to be softened is treated with calculated quantities of lime and soda at room temperature.
- Precipitates formed are finely divided.
- Small amounts of a **coagulant** such as **alum**, **aluminium sulphate**, **sodium aluminate** etc., are also added to flocculate.

Use of sodium aluminate as coagulant also helps the removal of silica as well as oil if present in water..

$$NaAlO_2 + 2H_2O \longrightarrow NaOH + Al(OH)_3$$

$$Al_2(SO_4)_3 + 3Ca(HCO_3)_2 \longrightarrow 2Al(OH)_3 + 3CaSO_4 + 6CO_2$$

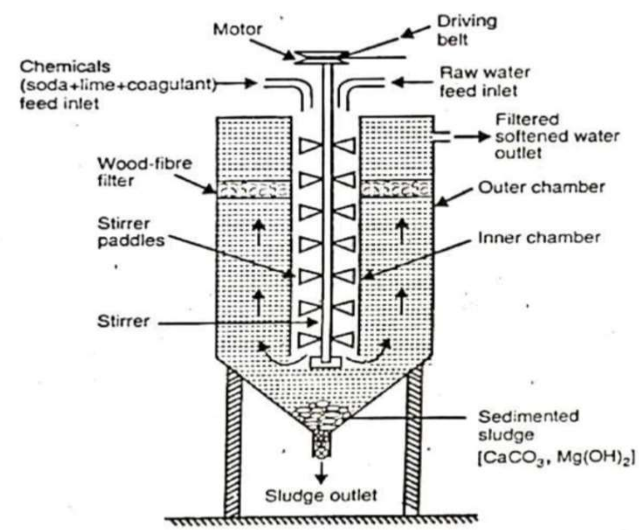


Fig. Continuous cold lime-soda softner

Sodium aluminate also helps in the removal of silica and oil present in water.

Now 100 parts by mass of $CaCO_3$ are equivalent to : (*i*) 74 parts of $Ca(OH)_2$, and (*ii*) 106 parts of Na_2CO_3.

$\therefore$ Lime requirement for softening

$$= \frac{74}{100}\left[\begin{array}{l}\text{Temp. } Ca^{2+} + 2 \times \text{Temp. } Mg^{2+} + \text{Perm. } (Mg^{2+} + Fe^{2+} + Al^{3+}) \\ + CO_2 + H^+ \text{ (HCl or } H_2SO_4) + HCO_3^- - NaAlO_2 \\ \text{all in terms of } CaCO_3 \text{ eq}\end{array}\right] \times \text{Vol. of water}$$

$\therefore$ Soda requirement for softening

$$= \frac{106}{100}\left[\begin{array}{l}\text{Perm. } (Ca^{2+} + Mg^{2+} + Al^{3+} + Fe^{2+}) + H^+ \text{ (HCl or } H_2SO_4) - HCO_3^- \\ \text{all in terms of } CaCO_3 \text{ eq}\end{array}\right]$$

$$\times \text{ Vol. of water}$$

Advantages:

1. Economical
2. Hot lime soda process is much **faster** than the cold lime soda process
3. During this process **pH value of water is increased** hence the corrosion of pipe is reduced
4. Besides the removal of hardness, the **quantity of minerals** in water is also **reduced**
5. Due to alkaline nature of water, amount of pathogenic bacteria in water are also removed
6. Requires **less amount of coagulants**

Disadvantages:

1. The softened water is **not completely free from hardness** (15-30ppm of hardness still remains)
2. Disposal of **large amount of sludge** is a problem
3. Careful operation and **skilled supervision** is required for efficient treatment of water

Difference between cold and hot lime soda process

Cold L-S Process	Hot L-S Process
Done at room temperature.	Done near the boiling point of water.
Reactions are slow and takes hours to complete.	Fast and takes only 15 minutes.
Precipitate is finely divided and therefore coagulants are needed.	Coarse precipitate coagulants not needed.
Tank size is large.	Small and compact.
Residual hardness 50 to 60 ppm.	15 to 30 ppm
Chemicals are consumed by dissolved gas,CO_2 etc.,	CO_2 and temporary hardness are automatically removed.

Numerical Problems Based on Lime-Soda Method

Example 1. *Calculate the amount of lime required for softening 50,000 litre of hard water containing $CaCO_3 = 25$ ppm, $MgCO_3 = 144$ ppm, $CaCl_2 = 111$ ppm, $MgCl_2 = 95$ ppm, $Na_2SO_4 = 15$ ppm $Fe_2O_3 = 25$ ppm.*

Solution. *Calculation of $CaCO_3$ equivalents :*

Constituent	*Multiplication factor*	*$CaCO_3$ equivalent*
$CaCO_3 = 25$ ppm	100/100	$25 \times 100/100 = 25.0$ mg/L
$MgCO_3 = 144$ ppm	100/84	$144 \times 100/84 = 171.43$ mg/L
$CaCl_2 = 111$ ppm	100/111	$111 \times 100/111 = 100.0$ mg/L
$MgCl_2 = 95$ ppm	100/95	$95 \times 100/95 = 100.0$ mg/L

$\therefore$ Lime requirement for softening

$$= \frac{74}{100}\left[\begin{array}{l}\text{Temp. } Ca^{2+} + 2 \times \text{Temp. } Mg^{2+} + \text{Perm. } (Mg^{2+} + Fe^{2+} + Al^{3+}) \\ + CO_2 + H^+ \text{ (HCl or } H_2SO_4) + HCO_3^- - NaAlO_2 \\ \text{all in terms of } CaCO_3 \text{ eq}\end{array}\right] \times \text{Vol. of water}$$

$\therefore$ Lime requirement for softening 50,000 L of hard water

$= 74/100\ [CaCO_3 + 2 \times MgCO_3 + MgCl_2 \text{ as } CaCO_3 \text{ eq}] \times \text{Vol. of water}$

$= 74/100\ [25.0 + 2 \times 171.43 + 100.0]$ mg/L $\times$ 50,000 L

$= 74/100\ [467.86$ mg/L$] \times 50{,}000$ L $= 1{,}73{,}10{,}820$ mg

$= (1{,}73{,}10{,}820/10^6)$ kg $=$ **17.311 kg.**

Example 2 *Calculate the quantity of lime and soda required for softening 50,000 litres of water containing the following salts per litre :* $Ca(HCO_3)_2 = 8.1$ *mg* ; $Mg(HCO_3)_2 = 7.5$ *mg* ; $CaSO_4 = 13.6$ *mg* ; $MgSO_4 = 12.0$ *mg* ; $MgCl_3 = 2.0$ *mg* ; and $NaCl = 4.7$ *mg*.

Solution. *Calculation of* $CaCO_3$ *equivalents :*

Constituent	*Multiplication factor*	$CaCO_3$ *equivalent*
$Ca(HCO_3)_2 = 8.1$ mg/L	100/162	$8.1 \times 100/162 = 5.0$ mg/L
$Mg(HCO_3)_2 = 7.5$ mg/L	100/146	$7.5 \times 100/146 = 5.14$ mg/L
$CaSO_4 = 13.6$ mg/L	100/136	$13.6 \times 100/136 = 5.0$ mg/L
$MgSO_4 = 12.0$ mg/L	100/120	$12.0 \times 100/120 = 10.0$ mg/L
$MgCl_2 = 2.0$ mg/L	100/95	$2.0 \times 100/95 = 2.11$ mg/L

Lime requirement for 50,000 L water

$$= 74/100\ [Ca(HCO_3)_2 + 2\ Mg(HCO_3)_2 + MgSO_4 + MgCl_2 \text{ as } CaCO_3 \text{ eq}] \times \text{Vol. of water}$$

$$= 74/100\ [5.0 + 2 \times 5.14 + 10.0 + 2.11]\ \text{mg/L} \times 50{,}000\ \text{L}$$

$$= 74/100\ [27.39\ \text{mg/L}] \times 50{,}000\ \text{L} = 10{,}13{,}430\ \text{mg}$$

$$= 10{,}13{,}430\ \text{mg} \times 10^{-6}\ \text{kg/mg} = \mathbf{1.0134\ kg}.$$

∴ Soda requirement for softening

$$= \frac{106}{100}\left[\begin{array}{l}\text{Perm. } (Ca^{2+} + Mg^{2+} + Al^{3+} + Fe^{2+}) + H^{+} \text{ (HCl or } H_2SO_4) - HCO_3^- \\ \text{all in terms of } CaCO_3 \text{ eq}\end{array}\right]$$

Soda requirement for 50,000 L water

$= 106/100\ [CaSO_4 + MgSO_4 + MgCl_2 \text{ as } CaCO_3 \text{ eq}] \times \text{Vol. of water}$

$= 106/100\ [10.0 + 10.0 + 2.11]\ mg/L \times 50{,}000\ L$

$= 106/100\ [22.11\ mg/L] \times 50{,}000\ L =$ 1171830 Mg

= 1.17 Kg

Example 3 . *A water sample contains the following impurities : $Ga^{2+} = 20$ ppm, $Mg^{2+} = 18$ ppm, $HCO_3^- = 183$ ppm and $SO_4^{2-} = 24$ ppm. Calculate the amount of lime and soda needed for softening.*

Solution. *Conversion into $CaCO_3$ equivalents :*

Constituent	*Multiplication factor*	*$CaCO_3$ equivalent*
$Ca^{2+} = 20$ ppm	100/40	$20 \times 100/40 = 50$ ppm
$Mg^{2+} = 18$ ppm	100/24	$18 \times 100/24 = 75$ ppm
$HCO_3^- = 183$ ppm	100/122	$183 \times 100/122 = 150$ ppm

Lime requirement $= 74/100\ [Mg^{2+} + HCO_3^-$ as $CaCO_3$ eq]

$= 74/100\ [75 + 150]$ ppm $=$ **166.5 ppm or mg/L.**

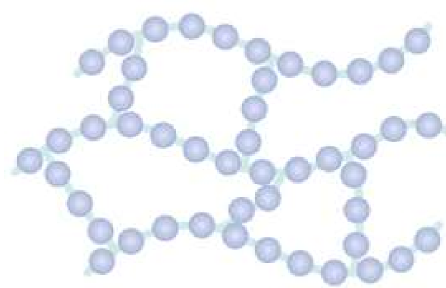
Soda requirement $= 106/100\ [Ca^{2+} + Mg^{2+} - HCO_2^-$ as $CaCO_3$ eq]

$= 106/100\ [50 + 75 - 150]$ ppm $=$ Negative or **nil.**

3. ION EXCHANGE/ DEMINERALIZATION PROCESS

- Cross linked long chain **organic polymers** with a **microporous structure**, and the **“functional Groups”** attached to the chains are responsible for the **ion-exchanging properties.**
- Resins with **acidic functional groups** are capable of exchanging **H^+** ions with other cations.
- Resins with **basic functional groups** are capable of exchanging **OH^-**ions with other anions.
- Resins are classified as:

i. Cation Exchange Resins ii. Anion Exchange Resins.



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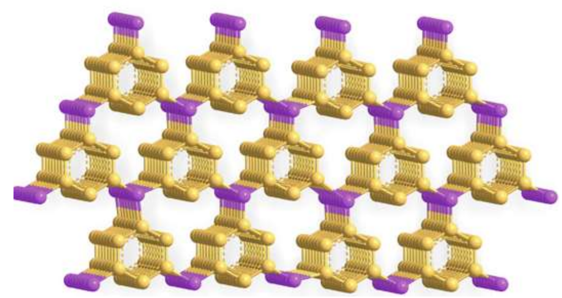


i. Cation Exchange Resins (RH^+) :

- Cation exchange resins are **styrene divinyl benzene co-polymers**.
- on sulphonation (or) carboxylation, which contains **–COOH**, **–SO_3H** functional groups (FG)
- These FG responsible for exchanging their **hydrogen ions** with **cations** in water.

$$2RH^+ + Ca^{2+} \longrightarrow R_2Ca^{2+} + 2H^+$$

$$2RH^+ + Mg^{2+} \longrightarrow R_2Mg^{2+} + 2H^+$$

(Exhausted Resin)

(RH^+ = Cation exchange)

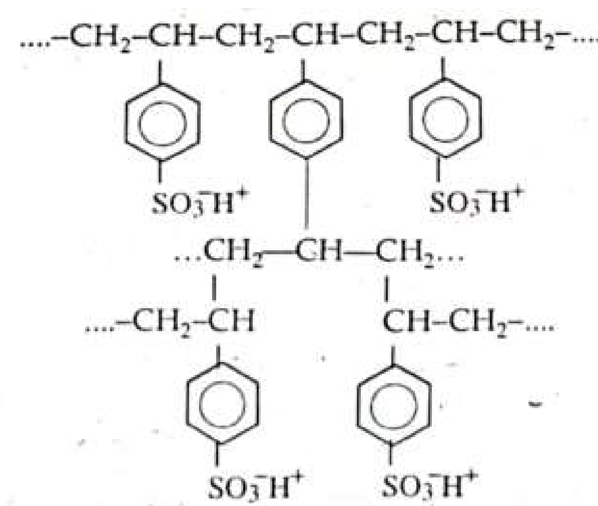


Fig. Acidic or cation exchange resin (sulphonate form)

ii. Anion Exchange Resins (OH^-):

- Anion exchange resins are **Phenol formaldehyde** (or) **amine formaldehyde copolymers**
- It contains **amino or basic functional groups** which responsible for exchanging their **OH^- ions** with anions in water.

$$R'OH^- + Cl^- \longrightarrow R'Cl^- + OH^-$$

$$2R'OH^- + SO_4^{2-} \longrightarrow R'_2SO_4^{2-} + 2OH^-$$

$$2R'OH^- + CO_3^{2-} \longrightarrow R'_2CO_3^{2-} + 2OH^-$$

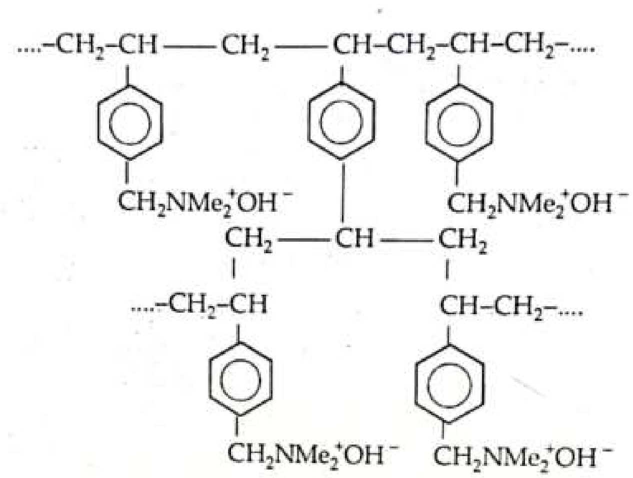


Fig. Basic or anion exchange resin (hydroxide form)

- **Cation exchange resins**, which **remove Ca^{+2} and Mg^{+2} ions** and **exchange an equivalent amount of H^+ ions**.
- **Anions exchange** resins remove **bicarbonates, chlorides and sulphates** from water exchange equivalent amount of **OH^- ions**.
- Thus by passing hard water through cation hardness is observed by the following reactions.
- **H^+ and OH^- ions**, thus released in water from respective **cation and anion exchange columns**, get combined to produce water molecules.

$$H^+ + OH^- \rightarrow H_2O$$

- **Softened water→ ion free from anions and cations→ low (2ppm) or zero hardness is obtained.**

Softening Cycle:

1. Cation Exchange Resin:

$$2RH^+ + Ca^{2+} \longrightarrow R_2Ca^{2+} + 2H^+$$
$$2RH^+ + Mg^{2+} \longrightarrow R_2Mg^{2+} + 2H^+$$

2. Anion Exchange Resin:

$$R'OH^- + Cl^- \longrightarrow R'Cl^- + OH^-$$
$$2R'OH^- + SO_4^{2-} \longrightarrow R'_2SO_4^{2-} + 2OH^-$$
$$2R'OH^- + CO_3^{2-} \longrightarrow R'_2CO_3^{2-} + 2OH^-$$

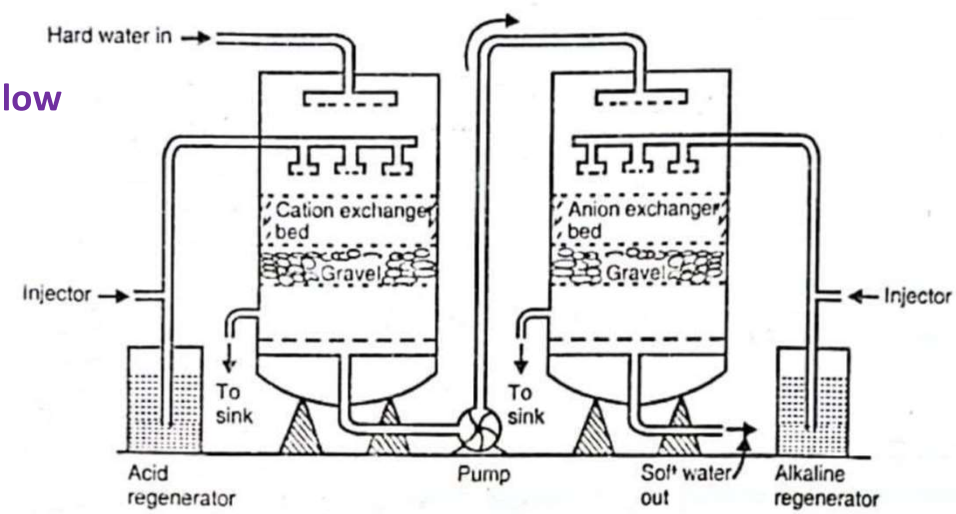


Fig. Ion Exchange Process

2. Regeneration Cycle:

- When cation exchanger losses capacity of producing **H^+ ions** and exchanger losses capacity of producing **OH^- ions**, they are said to be **exhausted**.

- The exhausted **cation exchanger** is regenerated by passing it through **dilute sulphuric acid**.

$$R_2Ca + 2HCl \rightarrow 2RH + CaCl_2$$
$$R_2Mg + 2H_2SO_4 \rightarrow 2RH + MgSO_4$$

$$R_2Ca^{2+} + 2H^+ \longrightarrow 2RH^+ + Ca^{2+} \text{ (washing)}$$

- The exhausted **anion exchanger** is regenerated by passing a **dilute solution of NaOH.**

$$R_2SO_4 + NaOH \rightarrow 2ROH + Na_2SO_4$$
$$RCl + NaOH \rightarrow ROH + NaCl$$
$$RHCO_3 + NaOH \rightarrow ROH + NaHCO_3$$

$$R'_2SO_4^{2-} + 2OH^- \longrightarrow 2R'OH^- + SO_4^{2-} \text{ (washing)}$$

Merits of the Ion exchange process:

- The process can be used to **soften highly acidic** or **alkaline water**.

- It produces water of very **low hardness (2ppm)**

- Very good for treating water for use in high-pressure boilers.

Demerits of Ion-exchange process:

- The **equipment is costly** and **more expensive chemicals** are needed.

- If water contains **turbidity**, the output of the process is reduced. (**Turbidity must be below 10ppm**; otherwise, it has to be removed by **coagulation** and **filtration**.)

Numerical Problems based on Ion Exchange Process:

Example 1: After treating 10^4 L of water by ion exchange, the cationic resin required 200 L of 0.1 N HCl & anionic resin required 200 L of 0.1 N NaOH solutions. Find hardness of the above sample of water.

Sol.

- **In an ion exchanger, all hardness causing cations are removed by cation exchanger; while anion exchanger removes anions of the constituents present in water.**
- **Consequently, the amount of acid used for generation of cation resin refers hardness of water.**

$\therefore$ Hardness in 10^4 L of water $\equiv 200$ L of 0.1 N-HCl $= 200$ L of 0.1 N-$CaCO_3$ eq

$= 200 \times 0.1$ L of 1 N-$CaCO_3$ eq $= 20$ L of 1N-$CaCO_3$

1N $CaCO_3 \equiv 50g$

$= 20 \times 50$ g of $CaCO_3$ eq $= 1{,}000$ g of $CaCO_3$ eq

$$\therefore \quad 1 \text{ hardness in } 1 \text{ L of water} \equiv \frac{1{,}000}{10^4} \text{ g of } CaCO_3 \text{ eq} = 0.1 \text{ g of } CaCO_3 \text{ eq}$$

$$= 0.1 \times 1{,}000 \text{ mg of } CaCO_3 \text{ eq} = 100 \text{ mg of } CaCO_3 \text{ eq}$$

Hence, hardness of water = **100 mg/L** or **ppm.**

Example 2:

1000 L water was softened by Ion Exchange method. For regeneration 150 L of 0.1 N each of HCl and NaOH was needed by respective exhaustive resins. Calculate the hardness of water.

Sol.

- **In an ion exchanger, all hardness causing cations are removed by cation exchanger; while anion exchanger removes anions of the constituents present in water.**
- **Consequently, the amount of acid used for generation of cation resin refers hardness of water.**

$\therefore$ Hardness in 1000 L of water $\equiv$ 150 L of 0.1 N-HCl = 150 L of 0.1 N-$CaCO_3$ eq

$= 150 \times 0.1$ L of 1 N-$CaCO_3$ eq = 15 L of 1N-$CaCO_3$

1N $CaCO_3$ $\equiv$ 50g

$= 15 \times 50$ g of $CaCO_3$ eq = 750 g of $CaCO_3$ eq

$\therefore$ 1 hardness in 1 L of water $\equiv \frac{750}{1000}$ g of $CaCO_3$ eq = 0.75 g of $CaCO_3$ eq

$= 0.75 \times 1{,}000$ mg of $CaCO_3$ eq = 750 mg of $CaCO_3$ eq

Hence, hardness of water = 750 mg/L or ppm.

Example 3: **After treating 50000 L of water by Ion Exchanger, the cationic rsin required 350 L of 0.5N HCl and anionic resin required 350 L of 0.5 N NaOH solution for regeneration. Calculate hardness of water sample.**

Solution:

In an ion-exchanger, all hardness causing cations are removed by cation exchanger while anion-exchanger removes anions of the constituents present in water. Hence, the amount of acid used for generation of cation resin refers to the hardness of water.

Total hardness of 50000 L water = 350 L of 0.5N HCl

= 350 L of 0.5N $CaCO_3$ equivalent

=(350*0.5)L of 1N $CaCO_3$ equivalent

= 175 L of 1 N $CaCO_3$ equivalent

= 8750 g of $CaCO_3$ equivalent

Hardness in 1 liters of water = 8750/ 50,000

= 0.175 g of $CaCO_3$ equivalent

Hardness of Water = 175 mg/L or ppm

4. DESALINATION OF BRACKISH WATER –REVERSE OSMOSIS

- The process of removing **common salt (Sodium Chloride)** from the water is known as **desalination**.

Depending upon the **quantity of dissolved salts**, water is graded as:

i. **Sea water** is about **35 parts** per thousand,ppt (35 parts in 10^3)

ii. **Brackish water** is between **0.5 and 30** parts per thousand.

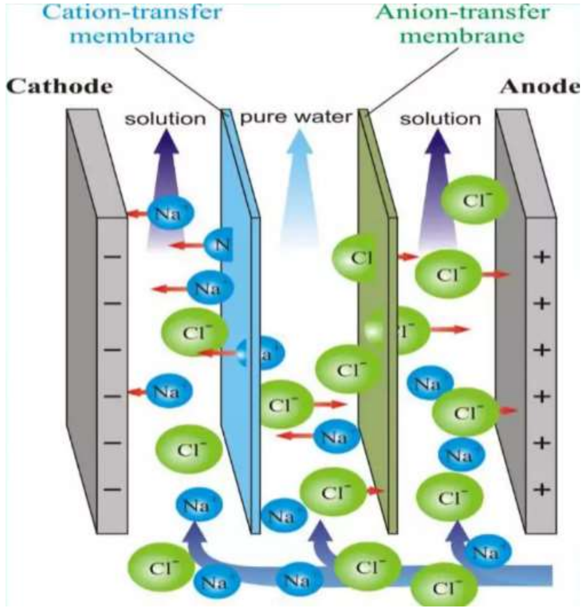
iii. **Fresh water** is less than **0.001 parts** per thousand.

- **Sea water and brackish** water can be made available as **drinking water** through the **desalination process.**

Desalination

Reverse Osmosis **Electrodialysis**

Electrodialysis Process



Components

- Power Source
- Cathode and Anode
- Ion Exchange Membranes

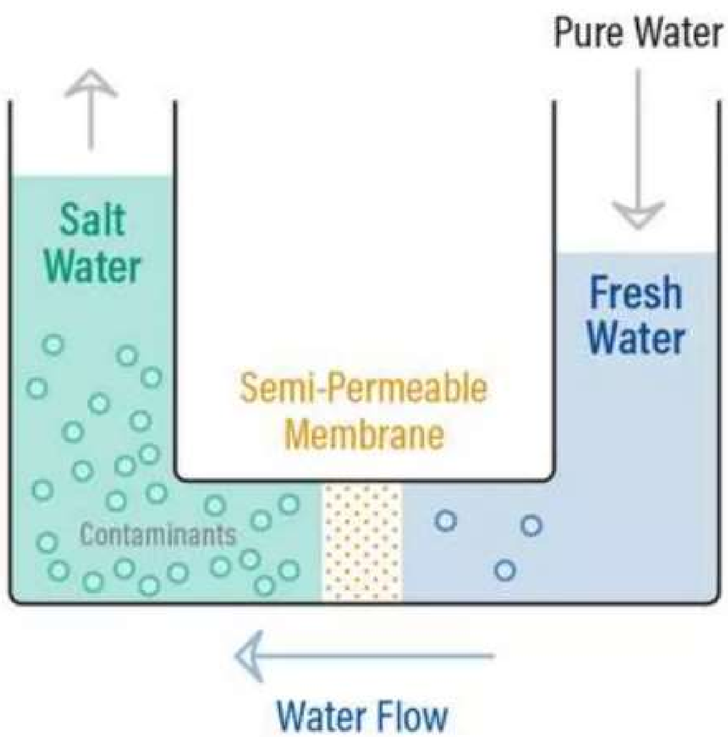
Electrodialysis process:

$$NaCl \xrightarrow{\text{Electric Current}} Na^{+} + Cl^{-}$$

Migrates towards Cathode (– ve)	***Migrates towards Anode (+ ve)***
Through cation selective membrane	***Through anion selective membrane***
(CSM, negatively charged membrane)	***(ASM, Positively charged membrane)***

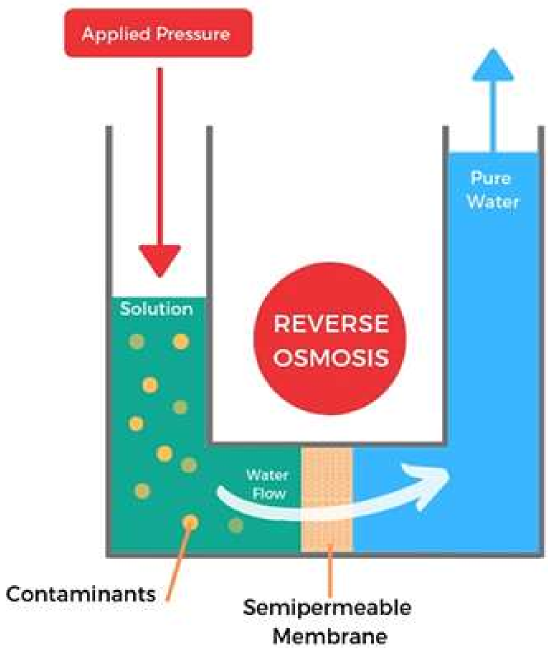
Osmosis:

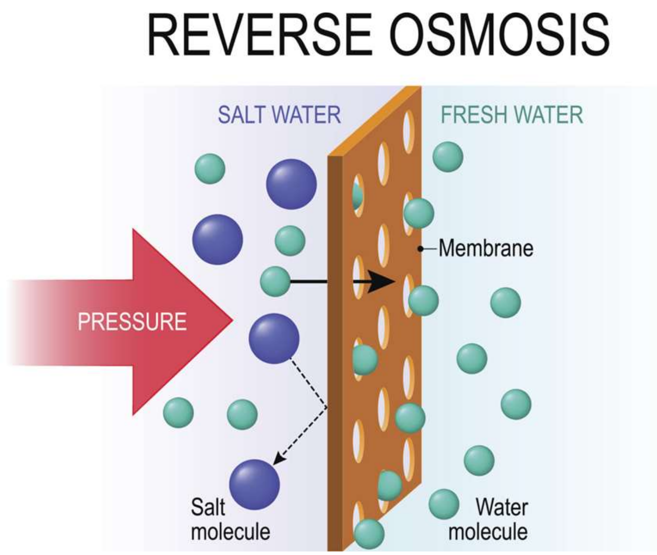
- When two solutions of **unequal concentrations** are separated by a **semi permeable membrane**, flow of solvent takes place from dilute to concentrate sides, due to osmosis.

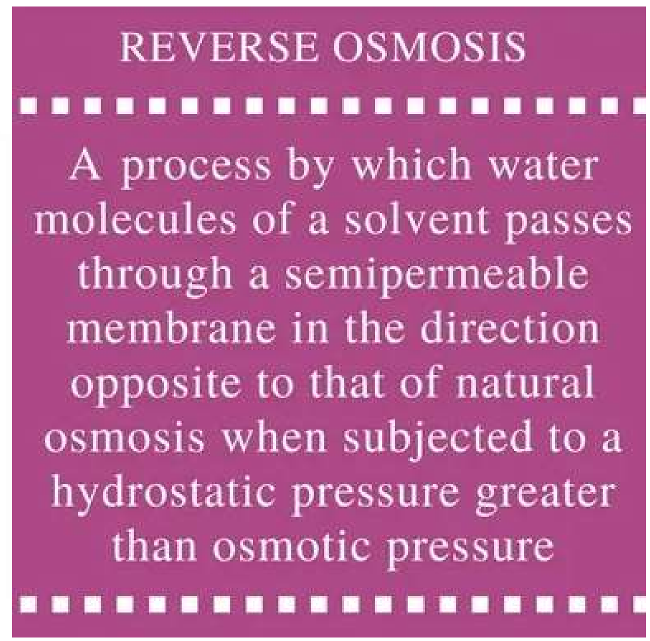


- **A solution that is less concentrated will have a natural tendency to migrate to a solution with a higher concentration**
- **A semi-permeable membrane is a membrane that will allow some atoms or molecules to pass but not others.**

- If, however a **hydrostatic pressure in excess to osmotic pressure** is applied on the concentrated side, the solvent flow is reversed, i.e, solvent is forced to move from concentrated side to dilute side across the membrane. This is the principle of **reverse osmosis (RO).**







How does Reverse Osmosis work?

- RO: In this method, pure solvent is separated from its contaminants, rather than removing contaminants from the water.
- The membrane filtration is sometimes also called **super-filtration** or **hyper-filtration**.

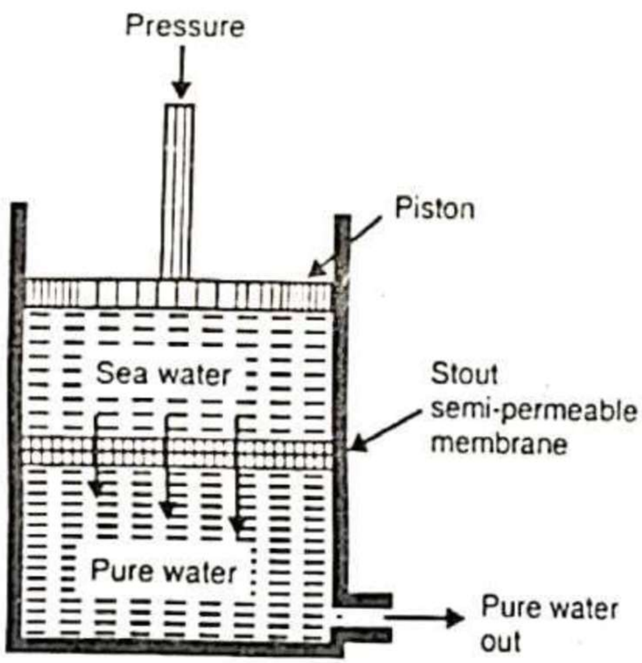
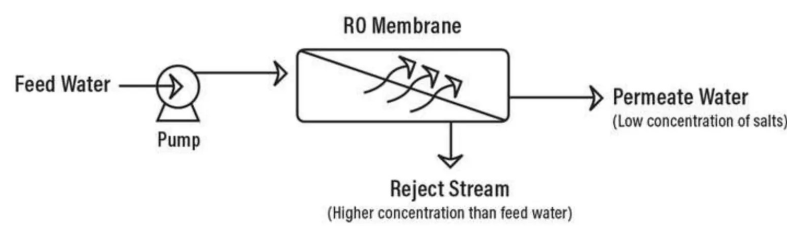


Fig: Reverse Osmosis Cell

Method:

- In this process, pressure is applied to the sea water or impure water to force the pure water content of it out the semipermeable membrane, **leaving behind the dissolve solids.**
- The membrane consists of very thin film of cellulose acetate, affixed to either side of a perforated tube.
- However, more recently **superior membranes** made of polymethacrylate and polyamide polymers have come into use.



Advantages:

1. RO possesses distinct advantages of **removing ionic** and **non-ionic**, **colloidal**, and **high molecular weight organic matter.**

2. It removes **colloidal silica**, which is not removed by **demineralization**.

3. The **maintenance cost** is almost entirely on the replacement of the **semi-permeable membrane.**

4. The **lifetime** of membrane is quite high, about **2 years**

5. The membrane can be **replaced within a few minutes**, thereby providing nearly uninterrupted water supply.

6. Due to **low capital cost**, **simplicity**, **low operating cost** and **high reliability**, the RO is gaining grounds at present for **converting sea water into drinking water** and for obtaining water for very **HP boilers**.

Disadvantages:

Reverse Osmosis wastes a lot of water

Reverse Osmosis removes all minerals

Dependency on electricity

The High Pressure Pump is required to extract Clear water from the membrane element.

Water taste altered